

Statistical methods for computer science

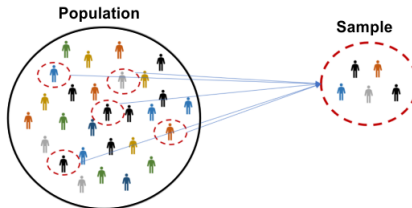
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Point Estimation

Goal of statistical inference

Inferential statistics

Draw conclusions about a **population** of interest based on **sample data**



We are interested in population characteristics and employ sample data to "infer" them

Which ingredients?

1. A **random variable** representing the phenomenon we are interested in
 - $Y =$ is a patient positive to covid19?
 - $Y =$ number of patients positive to covid19
 - $Y =$ costs to treat a patient positive to covid19?
2. A suitable **statistical model** for Y
 - $Y \sim Be(p)$
 - $Y \sim Po(\lambda)$
 - $Y \sim N(\mu, \sigma^2)$
 - $Y \sim ?(\mu, \sigma^2) \rightarrow$ partially specified model
3. The **sample data**, that is the realization of n iid random variables $Y_1, \dots, Y_n$, distributed according to the statistical model above

$$\mathbf{y} = (y_1, \dots, y_n)'$$

Which methods?

- Point estimation
- Interval estimation (confidence intervals)
- Hypothesis testing

Point estimation

Goal

Use sample data to make a guess about (i.e., **estimate**) the value of the parameters characterizing the model we chose for Y

- The target of inference are the parameters
 - μ
 - p
 - λ
 -
- The information we have comes from sample data $\mathbf{y} = (y_1, \dots, y_n)'$
- This are summarized by means of **appropriate statistics**

$$T(\mathbf{y}) = T(y_1, \dots, y_n)$$

- The goal thus become choosing the **most appropriate statistic** to estimate the parameters of interest

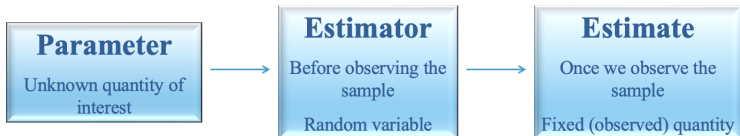
- When a sampling statistic is used for inferential purposes to **estimate** a parameter, it is called **estimator**
- Before observing the sample, the specific value of the characteristic of interest Y for the sampled units is unknown and so is the value of the estimator

$$Y_1, \dots, Y_n \rightarrow f(Y_1, \dots, Y_n)$$

- That is, the estimator is a **random variable**
- Once we observe the sample, we know the value of the characteristic Y for the sampled units and, in turn, we may compute the value that the estimator takes based on the observed sample

$$y_1, \dots, y_n \rightarrow f(y_1, \dots, y_n)$$

- This is known as **estimate** of the unknown parameter



In what follows, θ will be used to identify a generic parameter of interest, while $\hat{\theta}$ will be the corresponding estimator

Two questions

How to find (possibly good) estimators in practice?

- To answer the question, we need to delve into the different estimation methods (maximum likelihood, least squares, method of moments, etc...)
- Let's focus on **maximum likelihood estimation**

What is a good estimator?

- To answer the question, we need to identify desirable properties for the estimators

Maximum Likelihood Estimation (MLE)

- Estimate the parameter(s) of interest with the value(s) that maximize the probability (or the density) of observing the sample
- This probability is known as the **likelihood function**

Likelihood function

The **likelihood function** of a *fully specified model* is the joint probability of the sample data, considered as a function of the parameter(s)

$$L(\theta) = f(\mathbf{y} \mid \theta) = f(y_1, \dots, y_n \mid \theta)$$

Remarks

- Let's consider the joint distribution of the independent r.v.s $Y_1, \dots, Y_n$, distributed according to the pmf/pdf $f(\cdot | \theta)$

$$f(Y_1, \dots, Y_n | \theta) = \prod_{i=1}^n f(Y_i | \theta) \quad \text{due to independence} \quad (1)$$

- Before collecting the data, this is a function of the r.v.'s Y_i 's
- After we have collected the data, we assign to each Y_i its observed value y_i , and we can then consider equation (1) as a function of θ
- If we let θ vary,

$$L(\theta; y_1, \dots, y_n) = \prod_{i=1}^n f(y_i | \theta) \quad (2)$$

becomes a function that gives a *score* (or *likelihood*) to every possible value of θ

$$L(\theta; y_1, \dots, y_n) = L(\theta) = \prod_{i=1}^n f(y_i|\theta) \quad (3)$$

The **Maximum Likelihood Estimate (MLE)** of the parameter θ is the value, $\hat{\theta}$, that maximizes equation (3)

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

It is therefore a mathematical task

- Frequently, rather than dealing with the likelihood, we prefer its logarithmic transform (the **log-likelihood function**) as it generally has a simpler form

$$\ell(\theta) = \log L(\theta)$$

- We compute the first derivative of $\ell(\theta)$ wrt θ and equate it to zero

$$s(\theta) = \frac{\partial \ell(\theta)}{\partial \theta} = 0$$

- $s(\theta)$ is known as the **score function**
- We solve for θ and this leads to the MLE of θ

The Bernoulli case

Let $y_i \in \{0, 1\}$ be the observed values from a Bernoulli population $B(p)$.
The **likelihood function** is

$$\begin{aligned} L(p; y_1, \dots, y_n) &= p^{y_1} (1-p)^{1-y_1} \dots p^{y_n} (1-p)^{1-y_n} \\ &= p^{\sum_i y_i} (1-p)^{n-\sum_i y_i}, \quad y_i = 0, 1, \quad i = 1, \dots, n \end{aligned}$$

We take the logarithm and obtain the **log-likelihood function**

$$\ell(p; y_1, \dots, y_n) = \sum_i y_i \log(p) + (n - \sum_i y_i) \log(1-p)$$

$$\frac{\partial}{\partial p} \ell(p; y_1, \dots, y_n) = \frac{1}{p} \sum_{i=1}^n y_i - \frac{1}{1-p} \left(n - \sum_{i=1}^n y_i \right) = 0$$

$$\rightarrow \hat{p} = \frac{\sum_{i=1}^n y_i}{n}$$

Estimators and estimates of θ in a Bernoulli population

Then

$$\hat{p} = \frac{\sum_{i=1}^n y_i}{n}$$

is the MLE of p and

$$\hat{P} = \frac{\sum_{i=1}^n Y_i}{n}$$

is the corresponding (maximum likelihood) estimator

The first one is a (real) number, the second one is a r.v.

The Poisson case

Let $y_i \in \{0, 1, \dots\}$ be the observed values obtained from a population $Po(\lambda)$.

$$\begin{aligned} L(\lambda; y_1, \dots, y_n) &= \prod_{i=1}^n \frac{\lambda^{y_i} \exp[-\lambda]}{y_i!} \\ &= \frac{\lambda^{\sum_i y_i} \exp[-n\lambda]}{\prod_{i=1}^n y_i!} \end{aligned}$$

$$\ell(\lambda; y_1, \dots, y_n) = \sum_{i=1}^n y_i \log \lambda - n\lambda - \log c$$

$$\frac{\partial \ell(\lambda; y_1, \dots, y_n)}{\partial \lambda} = \frac{\sum_{i=1}^n y_i}{\lambda} - n = 0$$

$$\rightarrow \hat{\lambda} = \frac{\sum_{i=1}^n y_i}{n}$$

Estimators and estimates of θ in a Poisson population

Then

$$\hat{\lambda} = \frac{\sum_{i=1}^n y_i}{n}$$

is the MLE of λ and

$$\hat{\Lambda} = \frac{\sum_{i=1}^n Y_i}{n}$$

is the corresponding (maximum likelihood) estimator

The first one is a (real) number, the second one is a r.v.

The Gaussian case

Let $y_i \in \mathbb{R}, i = 1, \dots, n$, be the observed values obtained from a population $N(\mu, \sigma^2)$.

$$\begin{aligned} L(\mu, \sigma; y_1, \dots, y_n) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(y_i - \mu)^2}{2\sigma^2}\right\} \\ &= \left(\frac{1}{2\pi}\right)^{n/2} \frac{1}{\sigma^n} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right\} \end{aligned}$$

$$\ell(\mu, \sigma; y_1, \dots, y_n) = -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2$$

The mean

$$\frac{\partial}{\partial \mu} \ell(\mu, \sigma; y_1, \dots, y_n) = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \mu) = 0$$

$$\hat{\mu} = \bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

Estimators and estimates of μ in a Gaussian population

Then the ML estimates of μ is

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

while the corresponding estimator is

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$$

The variance

$$\frac{\partial}{\partial \sigma} \ell(\mu, \sigma; y_1, \dots, y_n) = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (y_i - \mu)^2 = 0$$

For **known** μ , the ML estimate of σ^2 is

$$s_{**}^2 = \frac{\sum_{i=1}^n (y_i - \mu)^2}{n}$$

For **unknown** μ , the ML estimate of σ^2 is

$$s_*^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n}$$

Estimators and estimates of σ^2 in a Gaussian population

The ML estimates of σ^2 is

$$s_{**}^2 = \frac{\sum_{i=1}^n (y_i - \mu)^2}{n} \quad \text{or} \quad s_*^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n}$$

while the corresponding estimators are

$$S_{**}^2 = \frac{\sum_{i=1}^n (Y_i - \mu)^2}{n} \quad \text{or} \quad S_*^2 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{n}$$

Evaluating a point estimator

Which are the desirable properties of an estimator?

- Unbiasedness
- Consistency
- Efficiency

Unbiasedness

An estimator $\hat{\theta}$ of a parameter θ is **unbiased** if its sampling distribution is centered around θ

$$E(\hat{\theta}) = \theta$$

or similarly

$$E(\hat{\theta}) - \theta = 0$$

- For a Bernoulli random variable

$$E(\hat{P}) = E\left(\frac{1}{n} \sum_i Y_i\right) = \frac{1}{n} \sum_i E(Y_i) = \frac{np}{n} = p$$

- For a Gaussian random variable

$$E(\hat{\mu}) = E(\bar{Y}) = E\left(\frac{1}{n} \sum_i Y_i\right) = \frac{1}{n} \sum_i E(Y_i) = \frac{n\mu}{n} = \mu$$

- An unbiased estimator on average provides the true value of θ
- On the contrary, an unbiased estimator tends to underestimate or overestimate the true θ

Consistency

An estimator $\hat{\theta}$ of a parameter θ is **consistent** if it converges in probability to θ as n increases; that is,

$$\Pr(|\hat{\theta} - \theta| \geq \epsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

In other words, its distribution gets closer to θ as n increases

- For a Gaussian random variable, the sample mean $\bar{Y}$ is consistent by the law of large numbers
- The same holds for the estimator $\hat{p}$ of the parameter p of a Bernoulli random variable

Efficiency

- Given two estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ of a parameter θ , the former is more efficient than the latter if it falls, on average, closer to θ
- A common way to evaluate this closeness is by means of the **Mean Squared Error (MSE)**

$$MSE(\hat{\theta}) = E_{\mathbf{Y}}(T(\mathbf{Y}) - \theta)^2$$

- For any estimator $\hat{\theta}$ a sufficient condition for consistency is that $MSE(\hat{\theta}) \rightarrow 0$ for $n \rightarrow \infty$.

Example

- Let $Y \sim Po(\lambda)$.
- Let's evaluate three estimators for λ

	$T(Y_1, \dots, Y_n)$	Note
T_1	$\frac{1}{n} \sum_{i=1}^n y_i$	makes sense
T_2	Y_1	loss of info
T_3	3	nonsense

Table: Three possible estimators for the parameter λ of a Poisson distribution

- Note that the parameter λ of a Poisson distribution is both the mean and the variance of the distribution.

- Let us evaluate the $MSE(\hat{\lambda})$ for $0 \leq \lambda \leq 5$. If we use the three estimators T_1 , T_2 , and T_3 from Tab. 1, we obtain

$$MSE(\hat{\lambda}_{T_1}) = E(\bar{Y} - \lambda)^2 = Var(\bar{Y}) = \frac{\lambda}{n}$$

$$MSE(\hat{\lambda}_{T_2}) = E(Y_1 - \lambda)^2 = Var(Y_1) = Var(Y) = \lambda$$

$$MSE(\hat{\lambda}_{T_3}) = E(3 - \lambda)^2 = (3 - \lambda)^2.$$

- T_3 is *observation free*.
- If $\lambda_1 \leq \lambda \leq \lambda_2$ the MSE of T_3 would be less than the MSE of the other two estimators, with $\lambda_{1,2} = \frac{6n+1}{n} \pm \frac{\sqrt{(\frac{6n+1}{n})^2 - 36}}{2}$
- The loss of T_1 is always less than loss of T_2

Back to efficiency

The MSE satisfies

$$\begin{aligned} \text{MSE}(\hat{\theta}) &= E_Y(\hat{\theta} - \theta)^2 \\ &= E \left[\left(\hat{\theta} - E(\hat{\theta}) \right) + \left(E(\hat{\theta}) - \theta \right) \right]^2 \\ &= E \left[\hat{\theta} - E(\hat{\theta}) \right]^2 + \left[E(\hat{\theta}) - \theta \right]^2 + 2 \left[E(\hat{\theta}) - E(\hat{\theta}) \right] \left[E(\hat{\theta}) - \theta \right] \\ &= E \left[\hat{\theta} - E(\hat{\theta}) \right]^2 + \left[E(\hat{\theta}) - \theta \right]^2 \\ &= \text{Var}[\hat{\theta}] + \text{bias}(\hat{\theta})^2 \end{aligned}$$

For an unbiased estimator $\hat{\theta}$ for a parameter θ , we have

$$\text{MSE}(\hat{\theta}) = \text{Var}[\hat{\theta}]$$

To wrap up

- For a Bernoulli random variable, the ML estimator of p is $\hat{P} = \bar{Y}$. This estimator is such that

$$\diamond \quad \text{bias}(\hat{P}) = E\left(\frac{1}{n} \sum_i Y_i\right) - p = \frac{1}{n} \sum_i E(Y_i) - p = \frac{np}{n} - p = 0$$

$$\begin{aligned} \diamond \quad \text{Var}[\hat{P}] &= \text{Var}\left(\frac{1}{n} \sum_i Y_i\right) = \frac{1}{n^2} \sum_i \text{Var}[Y_i] = \frac{np(1-p)}{n^2} \\ &= \frac{p(1-p)}{n} \quad \text{goes to 0 when } n \rightarrow \infty \end{aligned}$$

$$\diamond \quad \text{MSE}(\hat{P}) = \text{Var}[\hat{P}] = \frac{p(1-p)}{n}$$

$$\diamond \quad \hat{P} \sim \frac{\text{Bin}(n, p)}{n} \underset{n \rightarrow \infty}{\simeq} N\left(p, \frac{p(1-p)}{n}\right)$$

- For a Poisson random variable, the ML estimator of λ is $\hat{\Lambda} = \bar{Y}$. This is an estimator such that

$$\diamond \quad \text{bias}(\hat{\Lambda}) = E\left(\frac{1}{n} \sum_i Y_i\right) - \lambda = \frac{1}{n} \sum_i E(Y_i) - \lambda = \frac{n\lambda}{n} - \lambda = 0$$

$$\begin{aligned} \diamond \quad \text{Var}[\hat{\Lambda}] &= \text{Var}\left(\frac{1}{n} \sum_i Y_i\right) = \frac{1}{n^2} \sum_i \text{Var}[Y_i] = \frac{n\lambda}{n^2} \\ &= \frac{\lambda}{n} \quad \text{goes to 0 when } n \rightarrow \infty \end{aligned}$$

$$\diamond \quad \text{MSE}(\hat{\Lambda}) = \text{Var}[\hat{\Lambda}] = \frac{\lambda}{n}$$

$$\diamond \quad \hat{\Lambda} \sim \frac{\text{Po}(n\lambda)}{n} \underset{n \rightarrow \infty}{\simeq} N\left(\lambda, \frac{\lambda}{n}\right)$$

- For a Gaussian random variable, the ML estimator of μ is $\hat{\mu} = \bar{Y}$. This is an estimator such that

$$\diamond \quad \text{bias}(\bar{Y}) = E\left(\frac{1}{n} \sum_i Y_i\right) - \mu = \frac{1}{n} \sum_i E(Y_i) - \mu = \frac{n\mu}{n} - \mu = 0$$

$$\begin{aligned} \diamond \quad \text{Var}[\bar{Y}] &= \text{Var}\left(\frac{1}{n} \sum_i Y_i\right) = \frac{1}{n^2} \sum_i \text{Var}[Y_i] = \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \quad \text{goes to 0 when } n \rightarrow \infty \end{aligned}$$

$$\diamond \quad \text{MSE}(\bar{Y}) = \text{Var}[\bar{Y}] = \frac{\sigma^2}{n}$$

$$\diamond \quad \bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$